

Joseph Trevorrow

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EDUCATION

PhD Interactive AI, University Of Bristol

Sept 2023 – Sept 2027

Supervisors: Dr. Nirav Ajmeri (School of Computer Science), Dr. Emanuele Ratti (Dept. of Philosophy).

Provisional Thesis Title: “Value Aligned Multi-Agent Systems for Representative Decision Making”.

MRes Equivalent Training Year: Applied Data Science, Computational Logic, Human Computer Interaction (HCI), Natural Language processing (NLP), Responsible AI, and Uncertainty.

Research Interests: Multi-Agent Systems (MAS), Value Alignment, Civic AI, Socio-technical Systems, Computational Ethics, AI and Public/Industrial Policy.

BSc Hons. Computer Science, University of Liverpool

Sept 2020 – Jun 2023

First Class Honours. Focus: Multi-Agent Reinforcement Learning, AI.

Key Modules: High Performance Computing (in C and C++), Multi-Agent Systems, Vision.

Harington School, Uppingham Community College

Sept 2013 – Jun 2020

3 A Levels between A* and A, 11 GCSEs between grades 9-6.

PUBLICATIONS

Joseph Trevorrow, Nirav Ajmeri. “Hierarchical Consensus Value Aggregation for Decision Support”. In: *Proceedings of the 3rd International Workshop on Value Engineering in AI (VALE@ECAI)*. Springer Nature Switzerland, Oct. 2025, pp. 1–24.

- *Shortened Abstract:* Value aggregation is a method of group decision-making that combines agents’ personal values to ensure that decisions align to the beliefs and preferences of the group. Personal values represent an agent’s individual beliefs regarding what is important in their lives, motivating action. Current methods overlook agents’ preferences for different ethical principles, resulting in a consensus that may not accurately reflect the diverse perspectives within the group. We introduce Hierarchical Consensus Value Aggregation (HCVA), a novel method that elicits agents’ preferences of ethical principles. Using real-world data from the European Social Survey (ESS), we compare HCVA to competing aggregation techniques. Results show HCVA is better in representative and fairness metrics in situations where an agent’s principle preferences are representative of personal values.
- *Key Technical Skills Learned:* Value-Alignment and Decision-Making, Data Analysis, Philosophy and Politics, AI for Policymaking.
- *Key Soft Skills Learned:* Project management, Communication and Leadership of a project, Writing interdisciplinary work for a Computer Science audience.

EXPERIENCE

Teaching Assistant, University Of Bristol

Sept 2025 – Dec 2025

Lab supervision for a Year 3 elective unit (COMS30014 Artificial Intelligence) with ~100 students studying Prolog and MAS in the module. Assisted with marking and assessment.

Editor, The Inclusive Research Collective

Mar 2024 – May 2024

- Contributed guidance and edits to the “Inclusive Research Toolkit”, which aims to promote Equality, Diversity and Inclusion (EDI) in research and development.
- Worked as part of a team of cross-disciplinary editors to understand requirements for EDI in different fields of research. This resulted in more focused pages and accurate case studies for engineering researchers.

Programmer, ALT Group

June 2023 – Sept 2023

- Programmer in a small business (~ 3 employees) developing relevant AI applications for business technology (LeanIX).
- Directed AI guidance and decision making expertise when considering relevant Large Language Models (LLMs) during product planning phase using rapid prototyping.
- Gained significant insight into the needs of industry, from small businesses to enterprise systems, to give a full rounded perspective of business technology.

SKILLS

Expertise: Multi-Agent Systems, Reinforcement Learning, Ethics & Responsible AI, Rapid Prototyping, Analytical Skills, Technical Writing, Communication, Problem Solving, Python, Julia